The Primordial Photon Theory: Unification of All Matter via Strong Force Locking

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Abstract

This paper puts forward a new unified fundamental physical theory named the Primordial Photon Theory. Its core proposition holds that photons are the sole primordial origin of everything in the universe, and light exists as a pure energy carrier with no other forms of energy independent of light. This paper systematically sorts out all intrinsic properties of photons. Starting from the fundamental premise that photons possess no rest mass or kinetic energy and can only undergo contact interaction at the energy level, this paper refutes two long-standing traditional physical perceptions: wave-particle duality and the claim that light is electromagnetic waves. It also reinterprets three classic optical phenomena—thermal radiation induced by light, equivalent light pressure, and the photoelectric effect—in a self-consistent manner, clarifying that all three are secondary effects generated after photons unilaterally transfer energy, and thoroughly discards erroneous derivations involving mechanical collision and momentum transfer of photons.

This paper elaborates the generation mechanism of electric charge for protons and electrons: free photons carry transient virtual electromagnetic fields during

propagation. After ultra-high-energy gamma photons undergo contact interaction and rapid cooling and are confined within tiny microspaces via strong force locking, the virtual electromagnetic fields solidify into stable real electric charges. Combined with LHC heavy-ion and photon collision experimental data from CERN and astronomical observations of black hole jets, this paper verifies that ultra-high-energy gamma photons can only stably generate protons, neutrons and electrons after contact interaction and rapid cooling; newly formed particles will dissociate back into free photons instantly under high-temperature environments.

This paper provides a unified interpretation of the three fundamental cosmic forces, clarifying that the strong force, electromagnetic force and gravity all feature instantaneous remote action and attenuation following the inverse-square law without ever vanishing. A closed bidirectional cosmic cycle of photon-matter transformation dominated by gravity is constructed. This theory avoids logical contradictions arising from applying classical mechanics to massless photons and is compatible with existing collider observational results, offering a new self-consistent paradigm for high-energy physics and astrophysics.

Keywords:

Primordial Photon Theory; photon; contact interaction; strong force locking; refutation of wave-particle duality; photoelectric effect; LHC heavy-ion collision; three fundamental forces

1 Introduction

Modern high-energy physics, relying on the Large Hadron Collider (LHC) at the

European Organization for Nuclear Research (CERN) and various space astronomical observation instruments, has acquired substantial measured data on physical particles produced via contact interaction of high-energy photons, heavy ions and heavy nuclei. Astronomical observations of black hole jets have also confirmed that regions filled with pure high-energy photons in space gradually generate matter as the ambient temperature drops. Classical physics and quantum mechanics have long suffered internal logical inconsistencies when explaining the intrinsic nature of photons, wave-particle duality, the definition of electromagnetic waves, the photoelectric effect, light pressure and the origin of electric charge. They apply mechanical collision and momentum transfer models to massless photons and arbitrarily define photons as possessing both wave and particle properties, failing to provide a unified account of the origin of electric charge and to form a complete logical chain connecting the generation of micro-particles and cosmic evolution.

Based on laboratory measurements from CERN and astronomical observational evidence, this paper establishes the Primordial Photon Theory with the core axiom that light is the sole primordial energy of the universe. It revises various traditional misunderstandings of photons layer by layer and provides a unified explanation of optical phenomena, micro-particle generation, the nature of the three fundamental forces, and the entire cycle of cosmic matter and energy, constructing a new fundamental physical framework free of logical conflicts and consistent with all observed experimental phenomena.

2 The Full Spectrum of Photons

Visible light detectable by human eyes constitutes only a small segment of the full photon spectrum. Photons form a continuous spectrum with identical intrinsic nature and varying energy levels, ordered from low to high energy as follows: radio waves, microwaves, infrared light, visible light, ultraviolet light, X-rays, ordinary gamma photons, ultra-high-energy gamma photons.

Only ultra-high-energy gamma photons reach the energy threshold for matter creation, capable of generating protons, neutrons and electrons through contact interaction at the speed of light followed by cooling and strong force locking, which form all matter in the universe.

3 The Axiom of Constant Light Speed

There exist exactly three fundamental forces in the universe: the strong force, electromagnetic force and gravity, with no other forms of interaction. Each fundamental force operates under strict preconditions: the strong force and gravity only act on matter with mass, while the electromagnetic force only acts on charged particles.

Photons have no rest mass and carry no electric charge, so the three fundamental forces cannot pull, drag, accelerate or decelerate photons, and no force can alter their state of motion.

Photons maintain an inherent constant speed upon generation. Their intrinsic speed remains invariant whether in vacuum or media; only the propagation path changes after contact interaction, with no increase or decrease in speed.

4 All Intrinsic Properties of Photons

1. Zero volume, zero rest mass, zero electric charge

Photons exist as pure energy carriers without the volume, rest mass or charge of tangible matter, which is the fundamental reason why the three fundamental forces cannot act upon them. Photons are exempt from mechanical collisions in the classical sense and possess no kinetic energy. Collision and momentum impact described in classical mechanics do not apply to photons at all. Between photons, matter and electrons, only energy-level contact interaction occurs, which merely transfers energy without any mechanical collision or momentum exchange.

2. Rectilinear propagation with inherent vibration

Photons propagate in straight lines by default. During propagation, they maintain fixed vibration frequencies and vibration angles, which can be directly verified by polarizers.

3. Wave-particle duality is a false observational proposition

The intrinsic nature of photons is exclusively particle-based energy carriers with no inherent wave characteristics. The diffraction and fringe interference patterns observed in experiments are artificial artifacts formed when photons undergo contact interaction and trajectory scattering upon passing through slit walls. When no slit walls interfere with photons, they travel in straight lines exhibiting purely particle behavior with no wave manifestations. Wave-particle duality is a misinterpretation of observational phenomena rather than an inherent property of photons.

4. Refutation of the claim that light is electromagnetic waves, and clarification of

virtual electromagnetic fields

During linear propagation driven by inherent high-frequency vibration, photons generate transient virtual electric and magnetic fields. High-energy lasers produce detectable electromagnetic induction signals, while low-energy photons yield faint virtual field signals undetectable by conventional instruments. This phenomenon, however, does not equate photons to electromagnetic waves, as explained below: Classical electromagnetic waves rely on alternating electromagnetic fields as their carrier and driving force. In contrast, virtual electric and magnetic fields are secondary epiphenomena accompanying the linear motion of independent photon energy units, rather than the carrier or propulsive power of photons. Photons travel at a constant speed in straight lines even without such virtual electromagnetic effects.

Classical alternating electromagnetic fields can exist independently, while the virtual electromagnetic fields generated by photons cannot separate from photons. These virtual fields vanish instantaneously once photons are absorbed or stagnated. Radio waves, microwaves and other low-frequency radiations are fundamentally low-energy photons. Past research reversed cause and effect, deriving the false conclusion that light is electromagnetic waves solely based on its accompanying electromagnetic epiphenomena.

5. Unified underlying mechanism of thermal radiation, equivalent light pressure and photoelectric effect

Based on the rule that photons carry no mass or kinetic energy and only transfer energy via contact interaction, thermal radiation, equivalent light pressure and the

photoelectric effect are all secondary outcomes of unidirectional energy transfer from photons, with all mechanical collision interpretations discarded:

(1) The relationship between photons and heat

Heat arises from disordered vibration of micro-particles. When photons undergo contact interaction with atoms and electrons, they purely transfer energy to these particles, intensifying their disordered vibration and producing macroscopic temperature rise. Infrared photons readily transfer energy to micro-particles, leading to prominent thermal effects in daily life. Heat is a secondary outcome of intensified particle vibration after absorbing photon energy; it is not an inherent property carried by light waves.

(2) Correction of the traditional kinetic derivation of light pressure

Traditional theories explain light pressure via photon momentum and impact thrust, which is fundamentally erroneous. Massless photons possess no kinetic energy and cannot generate mechanical thrust.

The actual mechanism: sustained contact interaction between massive photons and the surface of matter continuously transfers energy to surface atoms, amplifying micro-vibration and outward expansion of atoms, creating an equivalent outward force measurable by instruments, defined as light pressure. Light pressure is merely a secondary equivalent effect of violent atomic vibration rather than a result of photon impact or thrust. Low-energy photons transfer trivial energy and produce undetectable equivalent light pressure, while high-energy lasers deliver massive continuous energy to generate obvious measurable light pressure.

(3) Underlying logic of the photoelectric effect, refuting both wave and collision hypotheses

Both the classical wave hypothesis and photon collision-momentum hypothesis fail to explain the photoelectric effect. Photons never collide with electrons.

Mechanism: Photons undergo contact interaction with internal electrons in metals and fully transfer their intrinsic energy to electrons. When the transferred energy exceeds the threshold binding electrons to atoms, electrons escape outward to form the photoelectric effect.

Core point: The entire process involves only pure energy transfer with no collision or momentum exchange. Low-energy photons cannot provide sufficient escape energy for individual electrons regardless of prolonged irradiation or expanded irradiation area, fully matching experimental observations without requiring wave or mechanical collision assumptions.

6. Virtual fields solidify into real electric charge after strong force locking After ultra-high-energy gamma photons undergo contact interaction and rapid cooling and are confined within the tiny interior space of protons and electrons, their transient virtual electromagnetic vibration fields are permanently locked and solidified into stable, permanent electric charge. Conclusion: The electric charge carried by protons and electrons originates entirely from the solidification of virtual electromagnetic fields of confined photons. Electric charge is not an inherent primitive property of micro-particles but a derivative produced when photons are confined via strong force locking.

7. Conditional superposition of photons Superposition of photons is not restricted by propagation directions. Photons moving opposite, intersecting or in parallel can produce superposition effects as long as their frequencies match. Ordinary daily light sources rarely satisfy the frequency matching condition to exhibit superposition.

8. Absorption, release and energy conversion of photons Photons can convert between high and low energy states. Electrons absorb and store photon energy via contact interaction and later release photons, forming micro-scale photon cycles.

5 Complete Self-Consistent Mechanism of Matter Generation

Ultra-high-energy gamma photons + contact interaction at light speed + rapid cooling + strong force locking = physical particles

1.Contact interaction of high-energy photons alone cannot form stable matter structures.

2.Rapid cooling is mandatory. Without cooling, newly formed particle structures melt and disintegrate instantly, reverting to free photons.

3.Strong force locking takes effect after cooling to permanently stabilize particle structures.

Protons, neutrons and electrons generated in this process constitute all atoms and macroscopic matter.

1 Dual Empirical Evidence: Astronomical Observation + Laboratory Experiments

1.Astronomical evidence: The proximal region of black hole jets contains only high-temperature pure gamma photons. As jets diffuse into space and cool down,

photons undergo contact interaction and locking to generate protons, neutrons and electrons in situ, demonstrating the natural transformation from photons to matter in the universe.

2.Laboratory evidence: Experiments of high-energy photon, heavy-ion and heavy-nucleus contact interaction at CERN stably produce fundamental particles, artificially replicating the full process of matter generation via photon contact interaction and cooling. The experiments are repeatable and provide tangible laboratory support for this theory.

It is fully proven that all matter consists of light energy sealed and solidified after cooling and strong force locking.

2 Two Forms of Macroscopic Matter

1.Direct atomic crystallization: Metals form via direct lattice stacking of atoms.

2.Molecular aggregation: Organic matter, liquids and gases form when atoms first bind into molecules via electromagnetic force then aggregate into macroscopic substances.

3.Rocks and soil are mixtures of atomic and molecular structures. Inorganic matter can restructure into organic matter; life is a highly ordered assembly of solidified light energy.

7 The Three Fundamental Cosmic Forces

There exist no fourth fundamental force in the universe. All micro, macroscopic and interstellar interactions belong to the three fundamental forces or their secondary derivatives.

Shared Universal Characteristics of the Three Forces

1.All act instantaneously over a distance with no propagation delay.

2.All attenuate following the inverse-square law of distance.

3.All weaken infinitely but never disappear completely.

4.Each force only manifests when its corresponding carrier exists: mass corresponds to strong force and gravity; electric charge corresponds to electromagnetic force.

5.Strong Force (Strong Force Locking Force) Carrier: Mass, effective only within the tiny scale of atomic nuclei. Core function: Lock nuclear structures formed by photon contact interaction, the fundamental guarantee for stable matter.

6.Electromagnetic Force Carrier: Electric charge, which originates from solidified virtual fields of confined photons. It governs atomic structures, molecular bonds, material mechanical properties, friction, all chemical reactions and the full process of photon absorption and emission by electrons.

7.Gravity (Chain Thrust Arising from Matter Evading Photons) Carrier: All massive matter, effective across the entire universe. Its essence is a chain drag effect generated when tangible matter evades free photons in space, maintaining galactic structures, stellar orbits and the overall evolution of the cosmos.

8 The Ultimate Cosmic Cycle Dominated by Gravity

1.Free high-energy photons solidify into fundamental particles via contact interaction, cooling and strong force locking, validated by CERN collider experiments and black hole jet observations.

2.Gravity accumulates matter and compresses it inward to build high temperature.

3.Extreme pressure inside black holes breaks the confinement of the strong force.

4.Solidified matter fully disintegrates, releasing all confined photons to revert into ultra-high-energy gamma photons ejected as jets.

5.Jets cool gradually while propagating outward, and photons undergo contact interaction and locking again to generate new matter.

The universe maintains an eternal cycle: light solidifies into matter, and matter decomposes back into light. Without gravity, stars and the complete energy cycle cannot form.

9 General Conclusions

1.The sole primordial origin of everything in the universe is light.

2.The root cause of constant light speed: Only the strong force, electromagnetic force and gravity exist in the universe. These three forces only act on matter with mass or electric charge. Photons carry no rest mass or charge and remain unaffected by any force, hence their speed is immutable.

3.Rules governing photon interaction: Photons have no rest mass or kinetic energy; mechanical collision does not exist. All interaction between photons and matter consists purely of energy-level contact interaction with zero momentum exchange.

4.Wave-particle duality is a false proposition. Observed fringe patterns arise from contact interaction and scattering between photons and slit walls, while photons possess only particle intrinsic nature.

5.Refutation of the claim that light is electromagnetic waves: Electric and magnetic

fields accompanying photon propagation are merely transient secondary epiphenomena, not the carrier or driving force of light, and epiphenomena cannot be equated with the intrinsic essence of light.

6.Unified law of photon-matter interaction: Thermal radiation, equivalent light pressure and the photoelectric effect are all secondary outcomes of unidirectional energy transfer from photons, with no photon impact or momentum transfer involved.

7.Origin of electric charge: Stable charge of protons and electrons forms when virtual electromagnetic fields of free photons solidify after confinement via strong force locking.

8.Rule of photon superposition: Photons of opposite, intersecting or parallel directions can superpose as long as their frequencies match.

9.Essence of matter: All tangible matter is light energy permanently sealed via strong force locking, mutually verified by astronomical observation of black hole jets and laboratory contact interaction experiments of photons, heavy ions and heavy nuclei at CERN.

10.System of cosmic fundamental forces: Only the strong force, electromagnetic force and gravity exist, acting instantaneously over distance and attenuating following the inverse-square law without vanishing.

11.Gravity is the core driving force of cosmic evolution, governing the infinite cyclic transformation between light and matter.

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